Ameet Rao's Math Notes

Tangent Planes, Critical Points, and the Second Partials Test

If f is differentiable at (x_0, y_0) Then $\nabla f(x_0, y_0)$ is normal to the level curve

Same fact as Theorem 13.12: if f is differentiable at (x_0, y_0), then $\nabla f(x_0, y_0)$ is normal to the level curve through that point. In 3D, ∇F is normal to a level surface, which is how tangent planes are built.

$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

$P:(x_0,y_0,z_0)$ $\quad n=\langle a,b,c\rangle$

$z^2-2x^2-2y^2=12$ $\quad P:(1,-1,4)$

$F(x,y,z)=z^2-2x^2-2y^2-12$

$F_x\big|_P=-4x$ $\quad F_y\big|_P=-4y$ $\quad F_z\big|_P=2z$

$=-4=a$ $\quad =4=b$ $\quad =8=c$

$n=\langle -4,4,8\rangle$ $\quad P:(1,-1,4)$

$-4(x-1)+4(y+1)+8(z-4)=0$

Tangent plane at P: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ with $\langle a, b, c\rangle = \nabla F(P)$. Write $F = z^2 - 2x^2 - 2y^2 - 12$; $F_x = -4x$, $F_y = -4y$, $F_z = 2z$ at $(1, -1, 4)$ give $n = \langle -4, 4, 8\rangle$, so $-4(x - 1) + 4(y + 1) + 8(z - 4) = 0$.

$z=1-\frac{1}{10}(x^2+y^2)$ $\quad (1,1,\frac{1}{2})$

$n=\langle -\frac{1}{5},-\frac{1}{5},-1\rangle$

$-\frac{1}{5}(x-1)-\frac{1}{5}(y-1)-1\left(z-\frac{1}{2}\right)=0$

For $z = f(x, y)$, use $F = f(x, y) - z$, so $n = \langle f_x, f_y, -1\rangle$. Heads-up: $(1, 1, ½)$ is not on $z = 1 - (x^2 + y^2)/10$, which gives $z = 4/5$ there. The textbook version $z = 1 - (x^2 + 4y^2)/10$ fits, with $n = \langle -1/5, -4/5, -1\rangle$.

If $f(x,y)$ has a local extrema then $f_x(a,b)=0$ and $f_y(a,b)=0$

(a,b) is a critical point

If f has a local max or min at (a, b), the surface is flat there: $f_x(a, b) = 0$ and $f_y(a, b) = 0$. Points where both partials are 0 (or one doesn't exist) are critical points, the candidates for extrema.

$f(x,y)=x^2+y^2-2x-6y+14$

$f_x=2x-2=0$

$f_y=2y-6=0$ $\quad \rightarrow (1,3)$

$f(x,y)=1-(x^2+y^2)^{1/3}$

$f_x=\frac{-2x}{3(x^2+y^2)^{2/3}}=0$ $\quad -2x=0$

$f_y=\frac{-2y}{3(x^2+y^2)^{2/3}}=0$ $\quad -2y=0$ $\quad \rightarrow (0,0)$

To find critical points, set both partials to 0 and solve together. For $x^2 + y^2 - 2x - 6y + 14$: $2x - 2 = 0$ and $2y - 6 = 0$ give $(1, 3)$. For $1 - (x^2 + y^2)^{(1/3)}$, the partials are undefined at $(0, 0)$, so it is critical too (a peak, $f = 1$).

2nd Partials Test

$d=\det\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy}\end{bmatrix}=f_{xx}f_{yy}-f_{yx}f_{xy}$

$d=f_{xx}f_{yy}-(f_{xy})^2$

$d=0$ $\quad$ Inconclusive

$d|_{(a,b)}>0$ and $f_{xx}(a,b)>0$ min

$d|_{(a,b)}>0$ and $f_{xx}(a,b)<0$ max

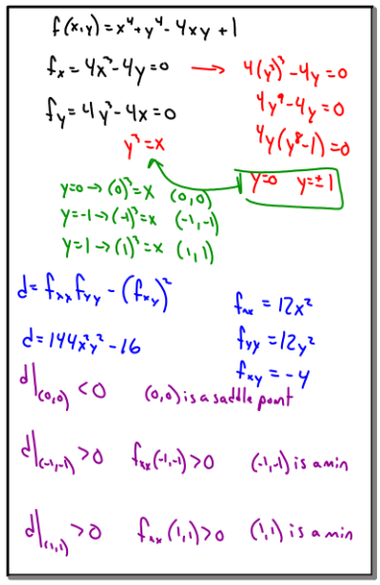
$d|_{(a,b)}<0$ $\quad$ Saddle point

Second Partials Test at a critical point (a, b): $d = f_{xx} \cdot f_{yy} - (f_{xy})^2$. $d > 0$ and $f_{xx} > 0$: relative min. $d > 0$ and $f_{xx} < 0$: relative max. $d < 0$: saddle point. $d = 0$: the test is inconclusive.

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Full process for f = x⁴ + y⁴ − 4xy + 1. Set f_y = 4y³ − 4x = 0, so x = y³, and put that into f_x = 4x³ − 4y = 0: 4y⁹ − 4y = 4y(y⁸ − 1) = 0, so y = 0, ±1, giving (0, 0), (1, 1) and (−1, −1). Then d = 144x²y² − 16: d(0, 0) = −16 < 0, a saddle point; at (1, 1) and (−1, −1), d = 128 > 0 and f_xx = 12 > 0, so both are relative minima with f = −1.